3.3. Tendo $p' = R \cdot p$, e $R = [r_1, r_2, r_3]$ sendo r_i os vetores coluna de R:

$$p' = r_1 \cdot p_1 + r_2 \cdot p_2 + r_3 \cdot p_3$$

Assim, $\sqrt{2} \cdot r_1 + 0 r_2 + 2 r_3 = \begin{bmatrix} 0 \\ 2 \\ \sqrt{2} \end{bmatrix} \rightarrow r_1 + \sqrt{2} r_3 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}$ (I)

$$r_1 + r_2 - r_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ -\sqrt{2} \end{bmatrix} \quad \text{(II)}$$

$$2\sqrt{2}\, r_2 = \begin{bmatrix} -\sqrt{2} \\ \sqrt{2} \\ -2 \end{bmatrix} \rightarrow r_2 = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} \quad \text{(III)}$$

(I) (II) $\rightarrow$ (III): $\begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix} - \sqrt{2}\, r_3 + \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} - r_3 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ -\sqrt{2} \end{bmatrix}$

$$(-\sqrt{2} - 1) r_3 = \begin{bmatrix} \frac{\sqrt{2}+1}{2} \\ \frac{-\sqrt{2}-1}{2} \\ -\frac{\sqrt{2}-2}{2} \end{bmatrix} \rightarrow r_3 = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$r_1 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix} - \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}$$

$$R = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{\sqrt{2}}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$